Investigating the Impact of Beer Tax on Fatalities Using Panel Regression

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Content:

- Introduction
- Data and Methodology
- Results
  - Random Effects Model
  - Fixed Effects Model
  - Statistical Tests
  - Policy Implications
- Potential Policy Implications

Introduction

This analysis explores the relationship between beer tax rates and fatalities across U.S. states over several years. Through panel regression techniques, we aim to estimate the effect of beer tax rates on fatalities and offer insights for public health policy.

Data and Methodology

Exploring structure and contents of the dataset and processing the data:

```
> head(Fatalities)
  state year spirits unemp   income   emppop  beertax baptist  mormon drinkage     dry youngdrivers    miles breath jail service fatal nfatal sfatal fatal1517
1    al 1982    1.37  14.4 10544.15 50.69204 1.539379 30.3557 0.32829    19.00 25.0063     0.211572 7233.887     no   no      no   839    146     99        53
2    al 1983    1.36  13.7 10732.80 52.14703 1.788991 30.3336 0.34341    19.00 22.9942     0.210768 7836.348     no   no      no   930    154     98        71
3    al 1984    1.32  11.1 11108.79 54.16809 1.714286 30.3115 0.35924    19.00 24.0426     0.211484 8262.990     no   no      no   932    165     94        49
4    al 1985    1.28   8.9 11332.63 55.27114 1.652542 30.2895 0.37579    19.67 23.6339     0.211140 8726.917     no   no      no   882    146     98        66
5    al 1986    1.23   9.8 11661.51 56.51450 1.609907 30.2674 0.39311    21.00 23.4647     0.213400 8952.854     no   no      no  1081    172    119        82
6    al 1987    1.18   7.8 11944.00 57.50988 1.560000 30.2453 0.41123    21.00 23.7924     0.215527 9166.302     no   no      no  1110    181    114        94
  nfatal1517 fatal1820 nfatal1820 fatal2124 nfatal2124  afatal     pop  pop1517  pop1820  pop2124 milestot unempus emppopus         gsp
1          9        99         34        120         32 309.438 3942002 208999.6 221553.4 290000.1    28516     9.7     57.8 -0.02212476
2          8       108         26        124         35 341.834 3960008 202000.1 219125.5 290000.2    31032     9.6     57.9  0.04655825
3          7       103         25        118         34 304.872 3988992 197000.0 216724.1 288000.2    32961     7.5     59.5  0.06279784
4          9       100         23        114         45 276.742 4021008 194999.7 214349.0 284000.3    35091     7.2     60.1  0.02748997
5         10       120         23        119         29 360.716 4049994 203999.9 212000.0 263000.3    36259     7.0     60.7  0.03214295
6         11       127         31        138         30 368.421 4082999 204999.8 208998.5 258999.8    37426     6.2     61.5  0.04897637
```

```
> summary(Fatalities)
     state       year       spirits          unemp            income          emppop          beertax           baptist           mormon           drinkage
 al     :  7   1982:48   Min.   :0.790   Min.   : 2.400   Min.   : 9514   Min.   :42.99   Min.   :0.04331   Min.   : 0.0000   Min.   : 0.1000   Min.   :18.00
 az     :  7   1983:48   1st Qu.:1.300   1st Qu.: 5.475   1st Qu.:12086   1st Qu.:57.69   1st Qu.:0.20885   1st Qu.: 0.6268   1st Qu.: 0.2722   1st Qu.:20.00
 ar     :  7   1984:48   Median :1.670   Median : 7.000   Median :13763   Median :61.36   Median :0.35259   Median : 1.7492   Median : 0.3931   Median :21.00
 ca     :  7   1985:48   Mean   :1.754   Mean   : 7.347   Mean   :13880   Mean   :60.81   Mean   :0.51326   Mean   : 7.1569   Mean   : 2.8019   Mean   :20.46
 co     :  7   1986:48   3rd Qu.:2.013   3rd Qu.: 8.900   3rd Qu.:15175   3rd Qu.:64.41   3rd Qu.:0.65157   3rd Qu.:13.1271   3rd Qu.: 0.6293   3rd Qu.:21.00
 ct     :  7   1987:48   Max.   :4.900   Max.   :18.000   Max.   :22193   Max.   :71.27   Max.   :2.72076   Max.   :30.3557   Max.   :65.9165   Max.   :21.00
 (Other):294   1988:48
      dry            youngdrivers          miles        breath      jail      service        fatal            nfatal            sfatal          fatal1517
 Min.   : 0.00000   Min.   :0.07314   Min.   : 4576   no :181   no  :241   no  :273   Min.   :  79.0   Min.   :  13.00   Min.   :  8.0   Min.   :  3.00
 1st Qu.: 0.00000   1st Qu.:0.17037   1st Qu.: 7183   yes:155   yes : 94   yes : 62   1st Qu.: 293.8   1st Qu.:  53.75   1st Qu.: 35.0   1st Qu.: 25.75
 Median : 0.08681   Median :0.18539   Median : 7796             NA's:  1   NA's:  1   Median : 701.0   Median : 135.00   Median : 81.0   Median : 49.00
 Mean   : 4.26707   Mean   :0.18593   Mean   : 7891                                    Mean   : 928.7   Mean   : 182.58   Mean   :109.9   Mean   : 62.61
 3rd Qu.: 2.42481   3rd Qu.:0.20219   3rd Qu.: 8504                                    3rd Qu.:1063.5   3rd Qu.: 212.00   3rd Qu.:131.0   3rd Qu.: 77.00
 Max.   :45.79210   Max.   :0.28163   Max.   :26148                                    Max.   :5504.0   Max.   :1049.00   Max.   :603.0   Max.   :318.00

   nfatal1517       fatal1820       nfatal1820       fatal2124       nfatal2124          afatal             pop               pop1517            pop1820
 Min.   : 0.00   Min.   :  7.0   Min.   :  0.00   Min.   : 12.0   Min.   :  1.00   Min.   :  24.6   Min.   :  479000   Min.   :  21000   Min.   :  21000
 1st Qu.: 4.00   1st Qu.: 38.0   1st Qu.: 11.00   1st Qu.: 42.0   1st Qu.: 13.00   1st Qu.:  90.5   1st Qu.: 1545251   1st Qu.:  71750   1st Qu.:  76962
 Median :10.00   Median : 82.0   Median : 24.00   Median : 97.5   Median : 30.00   Median : 211.6   Median : 3310503   Median : 163000   Median : 170982
 Mean   :12.26   Mean   :106.7   Mean   : 33.53   Mean   :126.9   Mean   : 41.38   Mean   : 293.3   Mean   : 4930272   Mean   : 230816   Mean   : 249090
 3rd Qu.:15.25   3rd Qu.:130.2   3rd Qu.: 44.00   3rd Qu.:150.5   3rd Qu.: 49.00   3rd Qu.: 364.0   3rd Qu.: 5751735   3rd Qu.: 270500   3rd Qu.: 308311
 Max.   :76.00   Max.   :601.0   Max.   :196.00   Max.   :770.0   Max.   :249.00   Max.   :2094.9   Max.   :28314028   Max.   :1172000   Max.   :1321004

    pop2124           milestot         unempus         emppopus          gsp
 Min.   :  30000   Min.   :  3993   Min.   :5.500   Min.   :57.80   Min.   :-0.123641
 1st Qu.: 103500   1st Qu.: 11692   1st Qu.:6.200   1st Qu.:57.90   1st Qu.: 0.001182
 Median : 241000   Median : 28484   Median :7.200   Median :60.10   Median : 0.032413
 Mean   : 336390   Mean   : 37101   Mean   :7.529   Mean   :59.97   Mean   : 0.025313
 3rd Qu.: 413000   3rd Qu.: 44140   3rd Qu.:9.600   3rd Qu.:61.50   3rd Qu.: 0.056501
 Max.   :1892998   Max.   :241575   Max.   :9.700   Max.   :62.30   Max.   : 0.142361
```

The data represents various socioeconomic factors and traffic accidents over a number of years. It contains statistics on fatalities, drunk driving, blood alcohol content (BAC), employment, population, and spending on spirits and beer.

Fatalities: The table contains statistics on fatalities likely related to traffic accidents. It shows minimum, 1st quartile, median, 3rd quartile, and maximum values for fatalities across different years.

Other factors: The table also includes various socioeconomic factors that might be related to traffic accidents. These include employment rates, population statistics, alcohol consumption measures (beer and spirits taxes), and driver demographics (young drivers).

Overall, the data seems to summarize relationships between traffic accidents (fatalities) and various socioeconomic factors over a period.

```
> # Q3: Preprocess the data as necessary, including handling missing values and converting variables to appropriate data types.
> colSums(is.na(Fatalities))
      state        year     spirits       unemp      income      emppop     beertax     baptist      mormon    drinkage         dry youngdrivers
          0           0           0           0           0           0           0           0           0           0           0           0
      miles      breath        jail     service       fatal      nfatal      sfatal   fatal1517  nfatal1517   fatal1820  nfatal1820   fatal2124
          0           0           1           1           0           0           0           0           0           0           0           0
  nfatal2124      afatal         pop     pop1517     pop1820     pop2124    milestot     unempus    emppopus         gsp
          0           0           0           0           0           0           0           0           0           0
> #Answer : there is no missing values in the beertax being the important explanatory varaible to explore
> # the relationship with fatalities. so there is no need to treat one missing values for jail and service.
> sum(is.na(Fatalities$beertax))
[1] 0
> sum(is.na(Fatalities$fatal_rate))
[1] 0
```

Before proceeding with the analysis, we conducted data preprocessing steps to ensure the integrity of the dataset. We first checked for missing values using the colSums(is.na(Fatalities)) function. We found that there were no missing values in the "beertax" variable, which is a crucial explanatory variable for exploring its relationship with fatalities. Therefore, there was no need to address the single missing value in the "jail" and "service" variables.

The "Fatalities" dataset from the "AER" package in R serves as the foundation for this analysis. Using the plm package, we employed panel regression models, incorporating random and fixed effects specifications.

Results

Random Effects Model

Significant variance in fatalities was attributed to individual-specific factors, with a standard deviation of 0.516. The estimated theta parameter indicated considerable heterogeneity.

Although the coefficient for beer tax was negative (-0.052), it was statistically insignificant at 5% significance level. (p-value = 0.6753)

The non-significant coefficient suggests that changes in beer tax rates do not significantly affect fatalities. Unobserved individual-specific factors may overshadow the direct impact of beer tax rates on public health outcomes.

Potential Policy Implications

Introducing higher beer tax rates presents a promising approach to curbing fatalities linked to alcohol consumption. Elevating the cost of alcoholic beverages may deter individuals from indulging in excessive drinking, potentially lowering the incidence of alcohol-related accidents and fatalities. Moreover, the increased revenue from raised beer taxes could be directed towards supporting public health endeavors and interventions aimed at mitigating alcohol-related harm.